MIT–Alpha Four-Rescale Dataset
Fixed and Dynamic Scheduling Described Separately

Data-Center Cooling Simulation Project

October 27, 2026

Abstract

One day of MIT Supercloud job GPU power is rescaled by 50, 100, 150 and 200, each with fixed and dynamic chiller scheduling, yielding eight 24 h Alpha cooling-plant simulations. All use the same supplied weather without weather shifts or extra artificial load scenarios. Only the two factor-50 runs are newly executed; the original six remain unchanged. Fixed and dynamic results are described separately, including mean power, daily electricity and trajectories for chillers, chilled-water pumps, condenser-water pumps and cooling-tower fans. Available individual-device powers are also exported. All eight runs complete and five pass every quality gate. Both factor-50 runs extrapolate the head MLP; R100_dynamic retains a short pump-efficiency violation.

1 Common source data and generation settings

The source day is August 31, 2021, 00:00–24:00 in `America/New_York`, corresponding to 04:00 UTC through 04:00 UTC the next day. The 86,400 one-second GPU records are averaged over 300 s intervals. Source mean power is approximately 30.348 kW and daily energy 728.360 kWh. These are job GPU-power records, not complete site IT electricity. Weather comes directly from the supplied 26-row historical reanalysis CSV, including adjacent boundaries; it is neither fetched anew nor modified. Hourly wet bulb is interpolated at five-minute centers and held within intervals, spanning approximately 15.713–20.258 °C.

With interval mean $\bar{P}_{\mathrm{GPU},k}$, requested cooling is

$$Q_{\mathrm{req},k} = \alpha \bar{P}_{\mathrm{GPU},k}, \qquad \alpha \in \{50, 100, 150, 200\}.$$

Rescaling changes amplitude only, preserving order and relative shape. It is a prescribed equivalent-load mapping, not PUE, a measured IT/GPU ratio or observed cooling demand. Model and dataset background is given in Refs. [1, 2, 3].

Table 1: Requested cooling in MW. Each factor is executed with fixed and dynamic policies, giving eight trajectories.

Factor	Mean	Minimum–maximum
50	1.517	1.342–1.687
100	3.035	2.685–3.373
150	4.552	4.027–5.060
200	6.070	5.369–6.746

The unmodified Alpha FMU is executed through FMI 2.0 Co-Simulation. Design temperature difference is 6 K and chiller supply setpoint 15 °C:

$$\dot{m} = \frac{Q_{\mathrm{req}}}{4180 \times 6}, \qquad T_{\mathrm{r}} = T_{\mathrm{s,true,\,previous}} + 6.$$

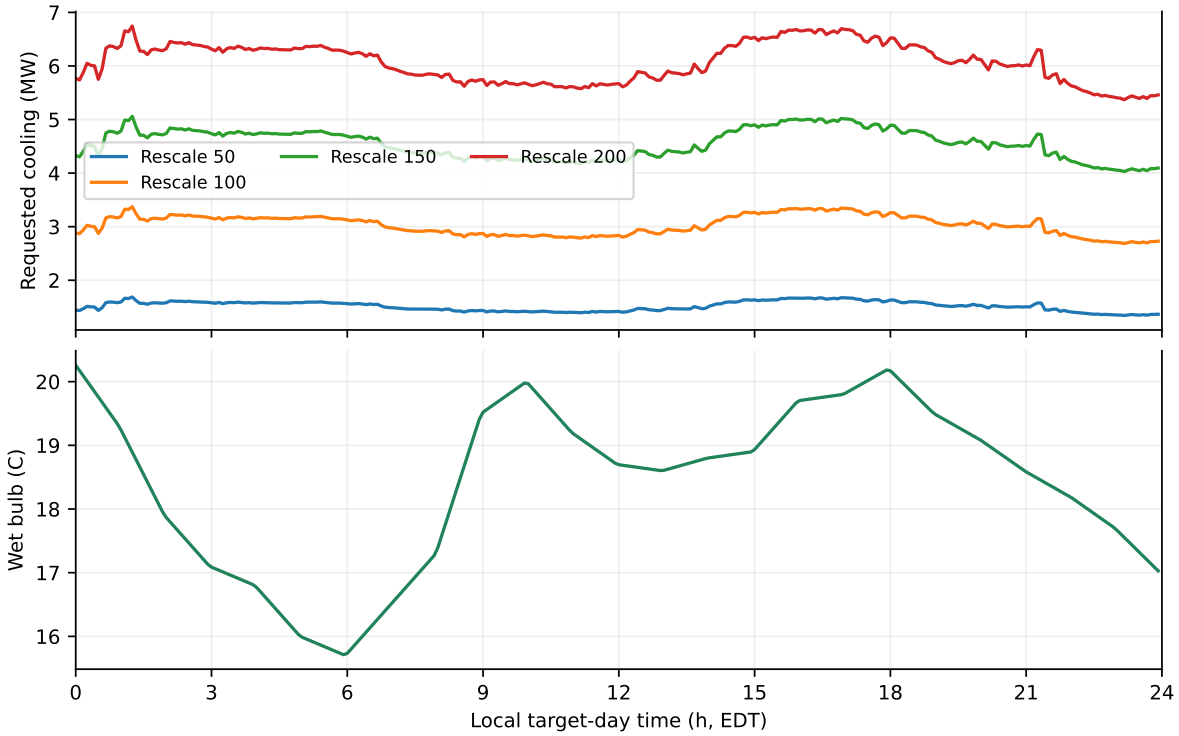


Figure 1: Four amplitude-rescaled requests and the same original weather used in all runs. No extra artificial scenarios are introduced.

Inputs are `Mchw` (kg/s), `Tchw_r` (K), and `Twb_outside` (K). Paired policies share requested cooling, mass flow and weather; return temperature follows each run's own true supply feedback. Load holds and decisions use 300 s intervals, supply feedback 30 s, FMI communication 10 s, and initialization tolerance 10^{-4}. These external intervals are distinct from the FMU's internal integration.

1.1 Warmup initialization and the start of delivered data

All eight runs first hold requested cooling and wet bulb at their respective first five-minute input values for 7200 s (2 h). Mass flow remains consistent with that initial cooling request and the 6 K design difference. Return temperature continues to follow the latest true supply temperature rather than freezing every water temperature. The assigned fixed or dynamic policy operates during warmup.

At the end of warmup, the supply-temperature range over the final 30 min must be at most 0.1 K, and the relative difference between mean plant powers in the two adjacent 15 min windows must be at most 1%. The protocol permits extension to 4/6 h if unstable; all eight actually pass these output-stability criteria at 2 h. This duration is an experimental setting rather than a universal minimum initialization time for Alpha.

The same FMU instance and continuous state are retained when the target-day time series begins; delivered relative time is then indexed from 0 s. Raw trajectories label initialization as `phase=warmup`. It is excluded from the delivered 24 h records, row counts, component electricity and COP. This computational preparation does not represent two preceding measured hours. Subsequent staging transients and MLP-extrapolation flags remain in the data.

2 Power components and statistical definitions

The FMU exposes four electrical-power categories. Plant total is their sum:

$$P_{\text{plant}} = P_{\text{chiller}} + P_{\text{CHWP}} + P_{\text{CDWP}} + P_{\text{CTfan}}.$$

Heat exchangers have no independent electrical-power output port; no zero or invented value is

Table 2: Native electrical-power outputs, in W; these are distinct from cooling demand.

Component	Category total	Individual-device output
Chillers	`P_Chillers_sum`	`Pchi1` through `Pchi6`, six units.
CHW pumps	`P_CHWPs_sum`	Individual pump powers are not exposed or artificially allocated.
CW pumps	`P_CDWPs_sum`	Individual pump powers are not exposed or artificially allocated.
Tower fans	`P_CTfans_sum`	`Pfan_CTi_01/02`, twelve fans across six towers.

assigned. Associated pump and fan electricity is included in the above categories.

Category 10 s means and energy use trapezoidal integration of the saved start/end powers; five-minute means aggregate interval energy. Reported daily mean power is energy averaged over the target 24 h (kW), while electricity is integrated energy (MWh). COP is delivered hydraulic heat divided by plant electricity, not PUE.

Individual chiller and fan powers are exported as 10 s and 300 s *endpoint values*, not interval means. The original records do not preserve each device's start value after every input update, so exact device-level interval electricity is not inferred. Category means, device endpoints and daily energy are named separately to prevent semantic confusion or double counting.

3 Fixed-scheduling results

The fixed policy retains the complete native two-chiller template, including linked pumps, valves and fans; MLP head is recomputed with boundaries. Constant count does not imply constant power. Y means all quality gates pass, while M identifies head-MLP input extrapolation.

Table 3: Fixed policy results. Electricity is in MWh.

Factor	Plant energy	COP	Count	Events	Quality
50	13.369	2.724	2	0	M
100	14.572	4.998	2	0	Y
150	16.746	6.524	2	0	Y
200	21.296	6.840	2	0	Y

Table 4: Fixed policy: daily mean component electrical power in kW. Totals are sums of the four categories; rounding may affect the last digit.

Factor	Chillers	CHW pumps	CW pumps	Tower fans	Plant total
50	321.24	46.43	169.22	20.18	557.06
100	366.16	52.99	167.83	20.18	607.16
150	457.49	59.33	160.75	20.18	697.74
200	639.12	65.40	162.66	20.18	887.35

Table 5: Fixed policy: daily component electricity in MWh.

Factor	Chillers	CHW pumps	CW pumps	Tower fans	Plant total
50	7.710	1.114	4.061	0.484	13.369
100	8.788	1.272	4.028	0.484	14.572
150	10.980	1.424	3.858	0.484	16.746
200	15.339	1.570	3.904	0.484	21.296

Factor 50 requests an average of approximately 1.517 MW but retains two chillers and their linked template, making pump/fan contributions particularly relevant. Its MLP inputs remain outside normalization bounds all day. Completion therefore does not establish validated operation at this joint low-load condition.

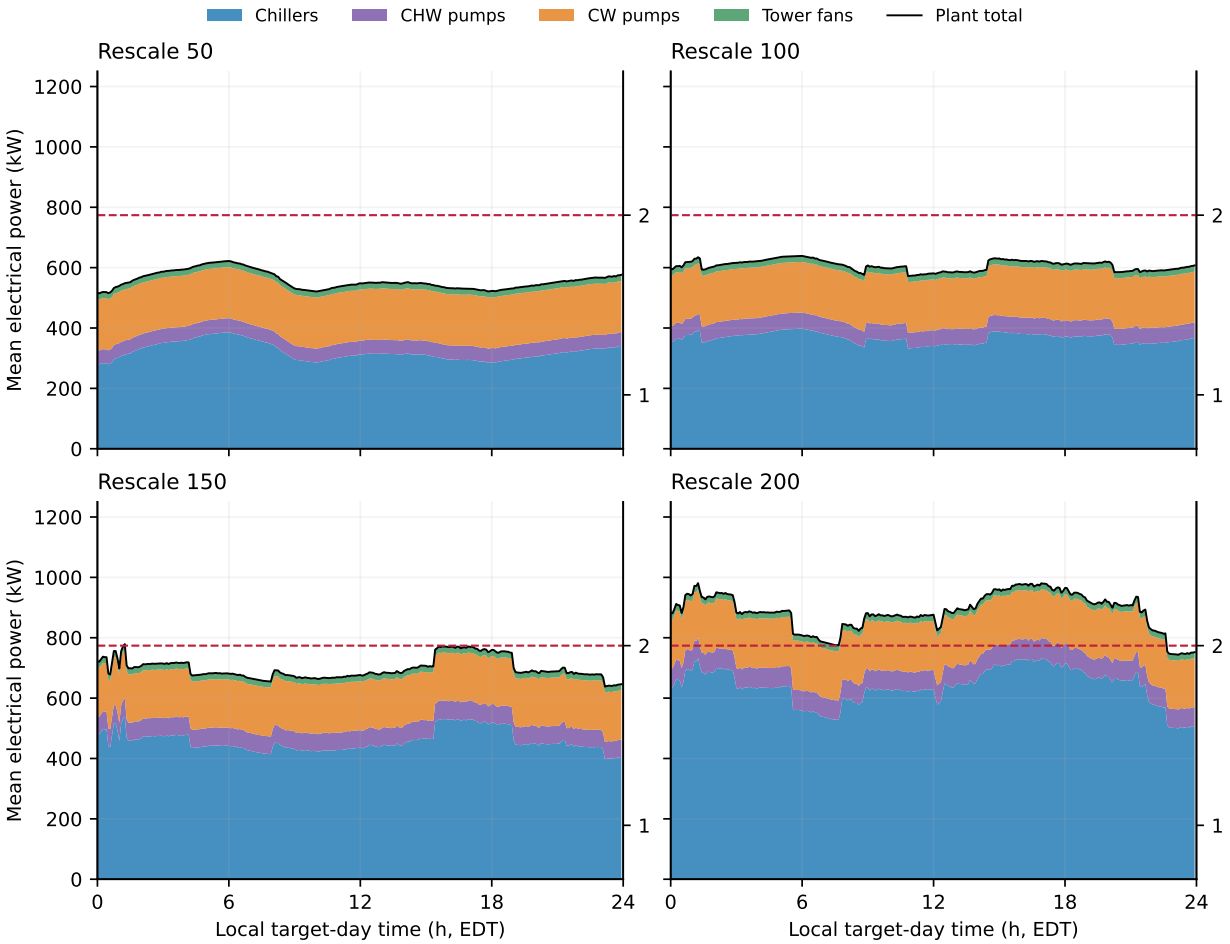


Figure 2: Fixed-policy results by factor. Stacked areas show four component five-minute mean powers, black shows plant total, and right-axis red dashed steps show active count, always two. Power scales match across panels. Factor 50 carries an MLP extrapolation flag.

4 Dynamic-scheduling results

The supervisor checks every 300 s. One-to-two staging requires load above 3.2222 MW; two-to-one requires load below 2.6180 MW. Conditions must persist for at least 600 s, with at least 1800 s since the previous configuration change. Initial count follows the initial request. Complete templates change together, potentially changing unit identities, pumps, valves and fans. Supplied weather keeps all runs in mechanical cooling (MC). The available one-/two-unit bank does not establish Alpha's maximum count or cooling capacity.

Table 6: Dynamic policy results. Electricity is in MWh; P indicates a pump-efficiency gate failure and M MLP input extrapolation.

Factor	Plant energy	COP	Count	Events	Quality
50	7.885	4.619	1	0	M
100	14.335	5.080	1–2	1	P
150	16.746	6.524	2	0	Y
200	21.296	6.840	2	0	Y

Table 7: Dynamic policy: daily mean component electrical power in kW.

Factor	Chillers	CHW pumps	CW pumps	Tower fans	Plant total
50	212.78	25.37	35.60	54.78	328.52
100	362.55	51.92	160.84	21.98	597.29
150	457.49	59.33	160.75	20.18	697.74
200	639.12	65.40	162.66	20.18	887.35

Table 8: Dynamic policy: daily component electricity in MWh.

Factor	Chillers	CHW pumps	CW pumps	Tower fans	Plant total
50	5.107	0.609	0.854	1.315	7.885
100	8.701	1.246	3.860	0.527	14.335
150	10.980	1.424	3.858	0.484	16.746
200	15.339	1.570	3.904	0.484	21.296

Factor 50 runs one unit throughout, with no target-day event. Factor 100 starts at one and naturally stages to two at 01:15 local time; no subsequent downward condition is met. Factors 150/200 start and remain at two, matching the fixed results; equality does not imply a disabled supervisor. Above $0.8nQ_n$ at the two-unit ceiling, the logged flag is staging-headroom information, not demonstrated unmet cooling.

Dynamic factor 50 also extrapolates the MLP throughout. Near the R100_dynamic upward event, two 10 s pump-efficiency outputs fall outside the allowed $[-0.01, 1.01]$ range, with minimum approximately -0.02319. Supply peaks at approximately 20.86 °C, and the longest continuous absolute error above 1 K is 30 s. All records and flags are retained; five-minute endpoint temperature alone can hide within-interval spikes.

Dynamic electricity changes relative to fixed by -41.03% for factor 50 and -1.63% for factor 100. Both dynamic runs fail at least one gate and change complete equipment configurations, so these differences establish neither validated savings nor an isolated count-only causal effect.

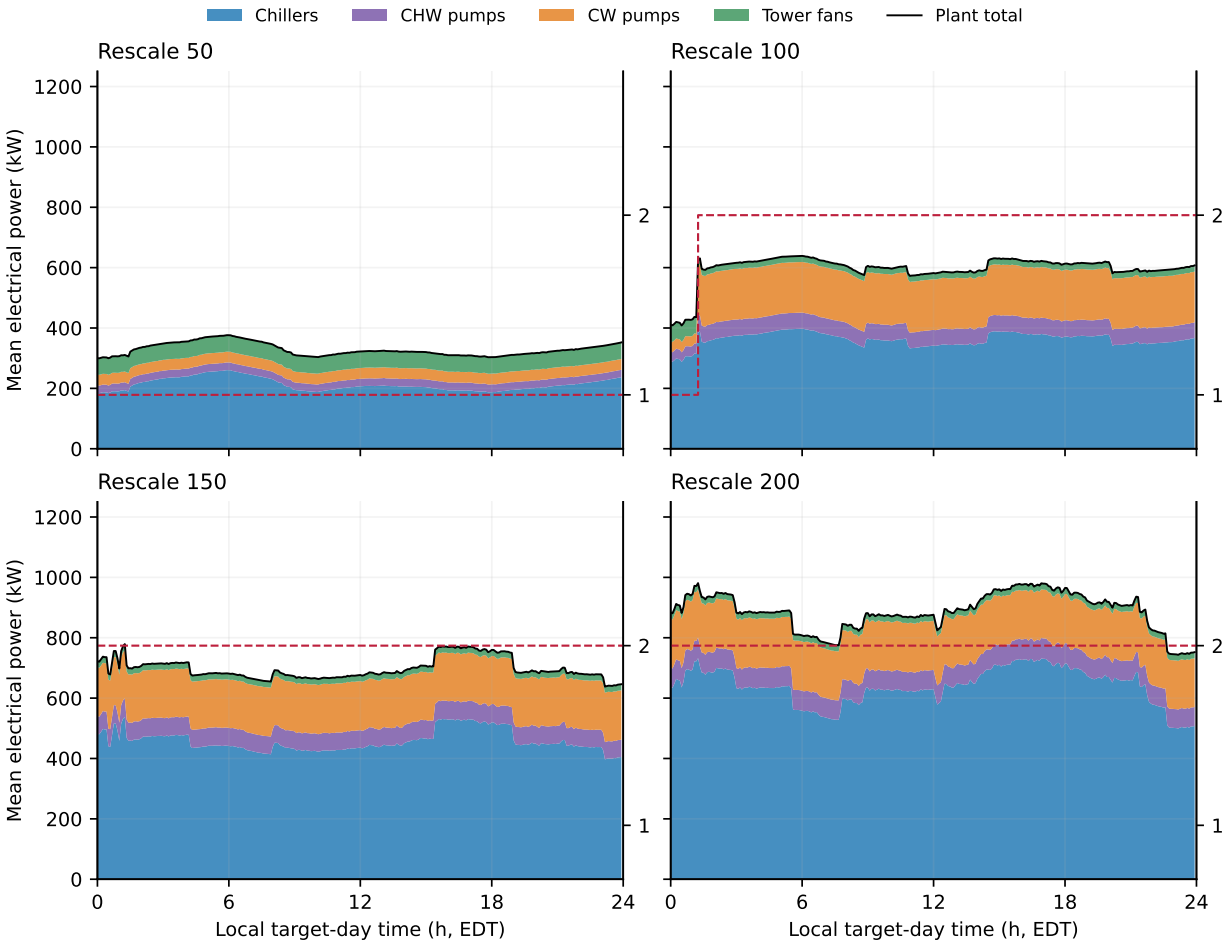


Figure 3: Dynamic-policy component powers and active count, with colors and scales matching the fixed-policy figure. Factor 50 retains one unit; factor 100 naturally stages once. They carry MLP-extrapolation and pump-efficiency quality flags, respectively.

5 Files, verification and limits

`simulated_300s.csv` contains 2,304 rows and `simulated_10s.csv.gz` contains 69,120. Public files are under `generated_data/`, including separate `fixed/` and `dynamic/` folders. This release distributes generated simulation outputs and this English description only. Original GPU/weather files, the 288-row source table, and the 1,152-row request-only table are not redistributed. Original GPU-power columns are also omitted from exported outputs; rescaled cooling requests and implemented boundary values remain as simulation context. Source access and attribution links are supplied separately. Experiments and quality flags are unchanged. Eight trajectories are not eight independent measured days.

File under generated_data/	Contents
`component_power_300s.csv`	Four component interval means and plant total, W.
`component_energy_summary.csv`	Per-run component daily mean kW, electricity MWh and endpoint peak kW.
`equipment_power_endpoints_10s.csv.gz`	Six chillers and twelve fans at 10 s endpoints, plus native category sums, W.
`equipment_power_endpoints_300s.csv`	The same native powers sampled at 300 s endpoints, W.
`run_summary.csv`, `comparison_by_rescale.csv`	Run quality and paired metrics; prose presents policies separately.
`control_decisions_300s.csv`, `switch_events.csv`	Supervisory decisions and the sole natural event.
`data_dictionary.csv`	Units and semantics for principal and individual-device output fields.

Temperatures are K, mass flow kg/s, and powers W; summary energy tables explicitly use MWh. Suffix `mean` denotes interval means and `start/end` denotes endpoints. `run_quality_pass` evaluates a whole run, with separate interval flags. Individual-device and category powers must not both be added again to plant total.

Gates include integrated absolute heat error at most 2%, five-minute absolute relative error P95 at most 5%, absolute supply error P95 at most 0.5 K, continuous error above 1 K at most 600 s, and MLP-bound, efficiency, finite-output, power and cooling-water checks. All eight complete and five pass; R50_fixed, R50_dynamic and R100_dynamic retain rejected flags.

Original production uses Juno job 479795; appended factor 50 uses 479840, with environment check 479839. Each batch is separately frozen, executed and downloaded; the assembled collection is not presented as an eight-case protocol frozen before the original runs. Both archive hashes match the remote values. Independent raw audits pass 568 and 264 checks, respectively, 832 in total; original six trajectory hashes remain unchanged. Published fields, component integrals, individual-power sums, policy-specific files and old/new consistency are checked separately. Evidence is in `verification/`; raw archive paths and hashes are in `metadata.json`.

These checks establish execution and accounting consistency, not field physical accuracy. The dataset has one GPU source day, no room thermal capacitance, and no calibration against measured MIT cooling outputs. MLP normalization bounds do not validate joint operating conditions. Derived trajectories from one source day must not be treated as independent dates in training/testing. Handle quality flags explicitly rather than silently deleting abnormalities and calling the result a complete normal trajectory.

References

[1] Siddharth Samsi, Matthew L. Weiss, David Bestor, et al. The MIT Supercloud dataset. arXiv:2108.02037, 2021.

[2] Si Wu, Wanfu Zheng, Zhe Wang, Guanghao Chen, Pu Yang, Shang Yue, Dingqian Li, and Yue Wu. AlphaDataCenterCooling: A virtual testbed for evaluating operational strategies in data center cooling plants. *Applied Energy*, 380:125100, 2025. doi: 10.1016/j.apenergy.2024.125100.

[3] Wanfu Zheng and contributors. AlphaDataCenterCooling source repository. https://github.com/wfzheng/AlphaDataCenterCooling, 2026. Commit 1f4a704cd8ff8c3f2679fd88b1d30bdec04275d2; accessed 2026-10-26.